

SIMATS ENGINEERING

ASSESSMENT TOOL 3 - Output Prediction

COURSE NAME: Data Structures

COURSE CODE: CSA0308

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Assessment Tool Weightage: 35%

COURSE OUTCOME COVERED

CO1: Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL3, BL4)

Questions: 5 **EACH QUESTIONS CARRIES:** 5 MARKS **Total Marks:** 25 **Duration:** ONE HOUR

S. No.	List of Questions
1	Array Traversal Program <pre>#include <stdio.h> int main() { int a[5] = {10, 20, 30, 40, 50}; int i; for(i = 0; i < 5; i++) printf("%d ", a[i]); return 0; }</pre>
2	Sum of Array Elements Program <pre>#include <stdio.h> int main() { int a[] = {2, 4, 6, 8}; int sum = 0, i; for(i = 0; i < 4; i++) sum += a[i]; printf("%d", sum); return 0; }</pre>



3	Recursive Factorial Program	<pre>#include <stdio.h> int fact(int n) { if(n == 0) return 1; return n * fact(n - 1); } int main() { printf("%d", fact(4)); return 0; }</pre>
4	Recursive Fibonacci Program	<pre>#include <stdio.h> int fib(int n) { if(n <= 1) return n; return fib(n-1) + fib(n-2); } int main() { printf("%d", fib(6)); return 0; }</pre>
5	Function as ADT Operation Program	<pre>#include <stdio.h> int add(int x, int y) { return x + y; } int main() { printf("%d", add(15, 25)); return 0; }</pre>

Code of Conduct:

I G RUPALAKSHMI(1911260197) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (5 Marks)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (3 Marks)	Level 1 – Beginning (1–2 Marks)
Understanding of Program Logic	1	Correctly identifies the complete program flow and execution sequence.	Minor misunderstanding of program flow.	Partial understanding of logic.	Unable to interpret the program logic.
Trace of Variables / Function Calls	1	Accurately traces all variable values, iterations, and recursive calls.	Traces most values correctly with minor errors.	Traces some values but misses important steps.	Incorrect or no tracing.
Output Prediction Accuracy	2	Output predicted completely and correctly.	Output mostly correct with minor mistakes.	Output partially correct.	Output incorrect or missing.
Complexity Interpretation (Time/Space)	1	Correctly identifies and justifies complexity where applicable.	Correct complexity with limited justification.	Partial understanding of complexity.	Incorrect or no complexity analysis.

1.PROGRAM CODING:

```
#include <stdio.h>

int main()
{
    int a[5] = {10, 20, 30, 40, 50};
    int i;

    for(i = 0; i < 5; i++)
    {
        printf("%d ", a[i]);
    }

    return 0;
}
```

OUTPUT:
10 20 30 40 50

CONCEPTS: TRAVERSAL OF ARRAY ELEMENTS

2.PROGRAM CODING:

```
#include <stdio.h>

int main()
{
    int a[] = {2, 4, 6, 8};
    int sum = 0, i;

    for(i = 0; i < 4; i++)
    {
        sum += a[i];
    }

    printf("%d", sum);

    return 0;
}
```

OUTPUT: 20
CONCEPTS: SUM OF ELEMENTS

3.PROGRAM CODING:

```
#include <stdio.h>

int fact(int n)
{
    if(n == 0)
        return 1;

    return n * fact(n - 1);
}

int main()
{
    printf("%d", fact(4));
    return 0;
}
```

OUTPUT:24

CONCEPT:FACTORIAL PROGRAM

4.PROGRAM CODING:

```
#include <stdio.h>
```

```
int fib(int n)
{
    if(n <= 1)
        return n;

    return fib(n - 1) + fib(n - 2);
}
```

```
int main()
{
    printf("%d", fib(6));
    return 0;
}
```

OUTPUT:8

CONCEPT:FIBONACCI PROGRAM

5.PROGRAM CODING:

```
#include <stdio.h>
```

```
int add(int x, int y)
{
    return x + y;
}
```

```
int main()
{
    printf("%d", add(15, 25));
    return 0;
}
```

OUTPUT:40

CONCEPT:ADDING USING A USER DEFINED FUNCTION